

Single Variable Calculus

Personal Notes

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1 Derivatives, Slope, Velocity, and Rate of Change

A derivative measures how fast a quantity changes at a point. On a graph, it gives the tangent's slope. For a moving object, it gives velocity. Both come from taking the limit of an average rate over a shrinking interval.

1.1 Secant Slopes and the Derivative

Let $y = f(x)$ and fix a point $P = (x_0, f(x_0))$ on the graph. A nearby point $Q = (x_0 + \Delta x, f(x_0 + \Delta x))$ determines a *secant line*. Its slope is the *difference quotient*

$$\frac{\Delta f}{\Delta x} = \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}.$$

As Q approaches P , the secant line may approach a limiting line: the tangent line at P .

Definition 1.1 [*Derivative*]. If the limit exists, the derivative of f at x_0 is

$$f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}.$$

The source states: “The derivative is the slope of the line tangent to the graph of $f(x)$ ” (printed p. 1; PDF p. 2). The limit makes this slope precise.

Meeting the graph at one point is not enough to define a tangent. A line can cross a curve once without following its direction. The limit of secant slopes describes the curve's direction near the point.

Note. The limit concerns nonzero values of Δx approaching zero. Substituting $\Delta x = 0$ into the difference quotient gives the meaningless expression $0/0$; we must simplify or otherwise analyze the quotient before taking its limit.

1.2 Example: the Reciprocal Function

Take $f(x) = 1/x$ at a point $x_0 \neq 0$. For $\Delta x \neq 0$,

$$\begin{aligned} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} &= \frac{1}{\Delta x} \left(\frac{1}{x_0 + \Delta x} - \frac{1}{x_0} \right) \\ &= \frac{1}{\Delta x} \left(\frac{-\Delta x}{x_0(x_0 + \Delta x)} \right) \\ &= -\frac{1}{x_0(x_0 + \Delta x)}. \end{aligned}$$

The cancellation makes the limit visible:

$$f'(x_0) = -\frac{1}{x_0^2}.$$

The negative sign agrees with the graph: $1/x$ is decreasing on either side of the origin.

The tangent line through $(x_0, 1/x_0)$ is therefore

$$y - \frac{1}{x_0} = -\frac{1}{x_0^2}(x - x_0).$$

Setting $y = 0$ gives the x -intercept $2x_0$. By symmetry of $y = 1/x$ under exchanging x and y , the y -intercept is $2/x_0$. The triangle cut off by the coordinate axes therefore has the constant area $\frac{1}{2}|2x_0||2/x_0| = 2$.

1.3 Derivative Notation

The derivative has several standard notations. Starting with

$$\Delta y = \Delta f = f(x_0 + \Delta x) - f(x_0),$$

the limiting relation can be written

$$\frac{\Delta y}{\Delta x} \longrightarrow \frac{dy}{dx} \quad \text{or} \quad \frac{\Delta f}{\Delta x} \longrightarrow f'(x_0).$$

Thus dy/dx , df/dx , f' , and Df all denote a derivative. The notation $f'(x_0)$ records the evaluation point explicitly; when using dy/dx , the point must be supplied by context.

1.4 The Power Rule from the Definition

For a positive integer n , let $f(x) = x^n$. The binomial expansion gives

$$(x + \Delta x)^n = x^n + nx^{n-1}\Delta x + (\Delta x)^2 R(x, \Delta x),$$

where R collects the remaining finitely many terms. Hence

$$\frac{(x + \Delta x)^n - x^n}{\Delta x} = nx^{n-1} + \Delta x R(x, \Delta x).$$

The second term vanishes as $\Delta x \rightarrow 0$, proving the *power rule* for monomials and, term by term, for polynomials:

$$\frac{d}{dx}x^n = nx^{n-1}.$$

1.5 Velocity as an Instantaneous Rate

The same difference quotient has a physical meaning. If $y(t)$ is the height of an object, then

$$\frac{\Delta y}{\Delta t}$$

is its average velocity over a time interval, while the limiting value $y'(t)$ is its *instantaneous velocity*.

For an object dropped from a height of 400 feet, the model

$$y(t) = 400 - 16t^2$$

places impact at $t = 5$ seconds because $y(5) = 0$. Its average downward speed over the full fall is $400/5 = 80$ feet per second. The instantaneous velocity is

$$y'(t) = -32t,$$

so at impact

$$y'(5) = -160 \text{ ft/s.}$$

The negative sign records direction: height is decreasing. Speed is the magnitude, 160 feet per second.

Note. The tangent-slope and velocity viewpoints are not separate definitions. Both are limits of average change over shorter and shorter intervals, which is why one derivative describes geometry and motion at once.

Source: [MIT 18.01 Single Variable Calculus](#), Fall 2006, Lecture 1, “Derivatives, Slope, Velocity, and Rate of Change,” taught by David Jerison.

2 Limits, Continuity, and Trigonometric Limits

Limits describe what a function approaches near a point. They let us define derivatives and continuity. We also need two trigonometric limits to find the derivatives of sine and cosine.

2.1 Rates of Change and Sensitivity

For related quantities x and $y = f(x)$, the quotient

$$\frac{\Delta y}{\Delta x}$$

measures average change over an interval. As $\Delta x \rightarrow 0$, it approaches the instantaneous rate dy/dx when the derivative exists. The same pattern appears in many settings:

$$\frac{dq}{dt} = \text{electric current}, \quad \frac{ds}{dt} = \text{velocity}, \quad \frac{dT}{dx} = \text{spatial temperature gradient}.$$

A derivative also measures *sensitivity*. If a measured quantity h determines an inferred quantity L , then $\Delta L/\Delta h$ describes how an error in h changes the calculated value of L , while dL/dh gives this sensitivity for very small changes. A large magnitude means that a small error in h can cause a much larger error in L .

2.2 Limits and Direct Substitution

Definition 2.1 [*Limit*]. We write

$$\lim_{x \rightarrow a} f(x) = L$$

when the values of $f(x)$ approach L as x approaches a through values $x \neq a$. The value $f(a)$ need not be used, or even defined.

Some limits are found by direct substitution. For example,

$$\lim_{x \rightarrow 3} \frac{x^2 + x}{x + 1} = \frac{3^2 + 3}{3 + 1} = 3.$$

This works because the expression is defined at $x = 3$ and its denominator stays nonzero nearby. In contrast, a derivative limit initially gives $0/0$ under direct substitution:

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}.$$

The fraction is considered only for $x \neq a$; it must be simplified or estimated before the limit is taken.

2.3 Continuity and One-Sided Limits

Definition 2.2 [*Continuity*]. The source says, “We say $f(x)$ is continuous at x_0 when”

$$\lim_{x \rightarrow x_0} f(x) = f(x_0)$$

(printed p. 3; PDF p. 4). This requires the value at the point and the limit to exist and agree.

Continuity says that the function’s nearby values agree with its assigned value at the point. To distinguish the two directions of approach, the *right-hand limit*

$$\lim_{x \rightarrow a^+} f(x)$$

uses only $x > a$, while the *left-hand limit* $\lim_{x \rightarrow a^-} f(x)$ uses only $x < a$. A two-sided limit exists exactly when the two one-sided limits exist and agree.

These limits separate four common failures of continuity.

- A *removable discontinuity* occurs when the two one-sided limits agree in a finite value L , but $f(a)$ is missing or differs from L . Defining $f(a) = L$ repairs the discontinuity. The function $\sin x/x$ at $x = 0$ will be an example once its limit is known.
- A *jump discontinuity* occurs when both one-sided limits are finite but unequal. Changing only $f(a)$ cannot join the two sides.
- An *infinite discontinuity* occurs when values become unbounded on at least one side, as $1/x$ does near zero.
- An *oscillatory discontinuity* occurs when values keep fluctuating without approaching any finite or infinite limit, as $\sin(1/x)$ does near zero.

2.4 Reading a Derivative Graph

The graph of a function and the graph of its derivative generally look quite different. For $f(x) = 1/x$,

$$f'(x) = -\frac{1}{x^2}.$$

The original function changes sign across the origin, whereas its derivative is negative on both sides. The derivative graph records slope, not height.

Symmetry also connects a function to its derivative. An *odd function* satisfies $f(-x) = -f(x)$; differentiating shows that f' is even. An *even function* satisfies $f(-x) = f(x)$; differentiating shows that f' is odd, whenever the derivatives exist.

For another picture, consider

$$y(t) = 400 - 16t^2, \quad -5 \leq t \leq 5.$$

It models an object rising from ground level, reaching its maximum height at $t = 0$, and returning to the ground. Its velocity

$$y'(t) = -32t$$

is positive during ascent, zero at the top, and negative during descent. The sign of the derivative therefore identifies the direction of motion directly.

2.5 Two Trigonometric Limits

Angles must be measured in *radians*. For $0 < \theta < \pi/2$, comparison of a triangle and a unit-circle sector gives

$$\sin \theta < \theta < \tan \theta.$$

Dividing through and rearranging yields

$$\cos \theta < \frac{\sin \theta}{\theta} < 1.$$

Both outer quantities approach 1 as $\theta \rightarrow 0^+$. Since $\sin \theta / \theta$ is even, the same conclusion holds from the left:

$$\boxed{\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1.}$$

The identity

$$1 - \cos \theta = \frac{(1 - \cos \theta)(1 + \cos \theta)}{1 + \cos \theta} = \frac{\sin^2 \theta}{1 + \cos \theta}$$

then gives

$$\frac{1 - \cos \theta}{\theta} = \frac{\sin \theta}{\theta} \frac{\sin \theta}{1 + \cos \theta} \rightarrow 1 \cdot 0,$$

so

$$\boxed{\lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta} = 0.}$$

2.6 Differentiability Implies Continuity

Theorem 2.3 [*Differentiable functions are continuous*]. If f is differentiable at a , then f is continuous at a .

Proof. For $x \neq a$, factor the change in f as

$$f(x) - f(a) = \frac{f(x) - f(a)}{x - a}(x - a).$$

As $x \rightarrow a$, differentiability makes the first factor approach $f'(a)$, while the second approaches 0. Their product therefore approaches 0, and hence $\lim_{x \rightarrow a} f(x) = f(a)$. $\square$

Note. The converse is false. The function $f(x) = |x|$ is continuous at zero but has different left- and right-hand slopes there, so it is not differentiable at zero.

Source: [MIT 18.01 Single Variable Calculus](#), Fall 2006, Lecture 2, “Limits, Continuity, and Trigonometric Limits,” taught by David Jerison.

3 Derivatives of Products, Quotients, Sine, and Cosine

We do not need to start from the derivative definition for every function. A few rules handle sums, products, quotients, sine, and cosine. The limits below explain why those rules work.

3.1 Linearity of the Derivative

Suppose u and v are differentiable functions of x , and let c be a constant. The *sum rule* and *constant-multiple rule* are

$$(u + v)' = u' + v', \quad (cu)' = cu'.$$

Both follow by splitting constants and sums inside the difference quotient. For the sum, for example,

$$\begin{aligned}(u + v)'(x) &= \lim_{\Delta x \rightarrow 0} \frac{u(x + \Delta x) + v(x + \Delta x) - u(x) - v(x)}{\Delta x} \\&= \lim_{\Delta x \rightarrow 0} \frac{u(x + \Delta x) - u(x)}{\Delta x} + \lim_{\Delta x \rightarrow 0} \frac{v(x + \Delta x) - v(x)}{\Delta x} \\&= u'(x) + v'(x).\end{aligned}$$

The calculation for cu is the same, with c factored outside the limit.

3.2 Derivatives of Sine and Cosine

Lecture 2 established the small-angle limits

$$\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1, \quad \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} = 0.$$

These formulas require angles in radians. Applying the sine addition identity to the derivative definition gives

$$\begin{aligned}\frac{d}{dx} \sin x &= \lim_{h \rightarrow 0} \frac{\sin(x + h) - \sin x}{h} \\&= \lim_{h \rightarrow 0} \left(\sin x \frac{\cos h - 1}{h} + \cos x \frac{\sin h}{h} \right) \\&= \cos x.\end{aligned}$$

A parallel calculation with $\cos(x + h) = \cos x \cos h - \sin x \sin h$ yields

$$\frac{d}{dx} \cos x = -\sin x.$$

The signs agree with the graphs: just to the right of zero, cosine decreases, while sine rises with slope close to one.

3.3 The Product Rule

The derivative of a product is not generally the product of the derivatives. Instead, the rule is

Theorem 3.1 [*Product rule*]. If u and v are differentiable, then (source, printed p. 2)

$$(uv)' = u'v + uv'.$$

Proof. Insert and subtract $u(x+h)v(x)$ in the numerator of the difference quotient:

$$\begin{aligned}(uv)'(x) &= \lim_{h \rightarrow 0} \frac{u(x+h)v(x+h) - u(x)v(x)}{h} \\ &= \lim_{h \rightarrow 0} \left[\frac{u(x+h) - u(x)}{h} v(x) + u(x+h) \frac{v(x+h) - v(x)}{h} \right].\end{aligned}$$

Differentiability implies continuity, so $u(x+h) \rightarrow u(x)$. Taking the limits of the two terms gives $u'(x)v(x) + u(x)v'(x)$. $\square$

There is also an area interpretation. If a rectangle has side lengths u and v , changes Δu and Δv change its area by

$$(u + \Delta u)(v + \Delta v) - uv = v \Delta u + u \Delta v + \Delta u \Delta v.$$

The first two terms are the two strips added along the original sides; the last is the small corner rectangle. Dividing by Δx gives

$$\frac{\Delta(uv)}{\Delta x} = v \frac{\Delta u}{\Delta x} + u \frac{\Delta v}{\Delta x} + \frac{\Delta u}{\Delta x} \Delta v.$$

As $\Delta x \rightarrow 0$, continuity gives $\Delta v \rightarrow 0$, so the corner contribution vanishes and the two strip contributions become $vu' + uv'$.

3.4 The Quotient Rule

For $v(x) \neq 0$, the rule is

Theorem 3.2 [*Quotient rule*]. If u and v are differentiable and $v \neq 0$, then (source, printed p. 4)

$$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}.$$

To see the numerator's order and the squared denominator, write $\Delta u = u(x + h) - u(x)$ and $\Delta v = v(x + h) - v(x)$. Then

$$\frac{u + \Delta u}{v + \Delta v} - \frac{u}{v} = \frac{v \Delta u - u \Delta v}{(v + \Delta v)v}.$$

Divide by h :

$$\frac{1}{h} \left(\frac{u + \Delta u}{v + \Delta v} - \frac{u}{v} \right) = \frac{v(\Delta u/h) - u(\Delta v/h)}{(v + \Delta v)v}.$$

As $h \rightarrow 0$, the two difference quotients approach u' and v' , while $\Delta v \rightarrow 0$. Therefore the right-hand side approaches $(u'v - uv')/v^2$.

Note. Both rules have two terms because both u and v can change. Swapping u and v leaves the product rule unchanged, but changes the quotient. For the quotient numerator, remember “derivative of the top times the bottom, minus the top times derivative of the bottom.”

Source: [MIT 18.01 Single Variable Calculus](#), Fall 2006, Lecture 3, “Derivatives of Products, Quotients, Sine, and Cosine,” presented by Kobi Kremnizer.

4 Chain Rule, and Higher Derivatives

The rules from Lecture 3 differentiate sums, products, and quotients. The next operation is *composition*: if $x = g(t)$ and $y = f(x)$, then

$$y = f(g(t)) = (f \circ g)(t).$$

The chain rule multiplies the rate of the outer function by the rate of the inner function.

4.1 The Chain Rule

Write the changes in the three variables as Δt , Δx , and Δy . When the relevant quotients are defined,

$$\frac{\Delta y}{\Delta t} = \frac{\Delta y}{\Delta x} \frac{\Delta x}{\Delta t}.$$

As $\Delta t \rightarrow 0$, continuity of g gives $\Delta x \rightarrow 0$. This suggests the derivative formula below.

Theorem 4.1 [*Chain rule*]. If g is differentiable at t and f is differentiable at $g(t)$, then

$$\frac{d}{dt}f(g(t)) = f'(g(t))g'(t).$$

Equivalently, if $y = f(x)$ and $x = g(t)$, then

$$\frac{dy}{dt} = \frac{dy}{dx} \frac{dx}{dt}.$$

Proof. Set $x_0 = g(t)$ and let $\Delta x = g(t + h) - g(t)$. Differentiability of f at x_0 means that

$$f(x_0 + \Delta x) - f(x_0) = f'(x_0)\Delta x + r(\Delta x)\Delta x,$$

where $r(\Delta x) \rightarrow 0$ as $\Delta x \rightarrow 0$; set $r(0) = 0$. Dividing by h gives

$$\frac{f(g(t + h)) - f(g(t))}{h} = (f'(x_0) + r(\Delta x)) \frac{g(t + h) - g(t)}{h}.$$

Letting $h \rightarrow 0$ yields $f'(g(t))g'(t)$. □

Example: Let $f(x) = \sin x$ and $g(t) = t^2$. Then

$$\frac{d}{dt} \sin(t^2) = \cos(t^2) \cdot 2t = 2t \cos(t^2).$$

Differentiate the outer sine to get cosine, keep its input t^2 , then multiply by the derivative $2t$ of that input.

Order matters in composition. With $f(x) = \sin x$ and $g(x) = x^2$,

$$(f \circ g)(x) = \sin(x^2), \quad (g \circ f)(x) = \sin^2 x.$$

Thus $f \circ g$ and $g \circ f$ need not be equal.

4.2 Two More Chain-Rule Examples

For $y = \cos(1/x)$ with $x \neq 0$, set $u = 1/x$. Since

$$\frac{dy}{du} = -\sin u, \quad \frac{du}{dx} = -\frac{1}{x^2},$$

the two negative signs cancel:

$$\frac{d}{dx} \cos\left(\frac{1}{x}\right) = \frac{\sin(1/x)}{x^2}.$$

The chain rule also extends the power rule to negative integer exponents. For a positive integer n ,

$$x^{-n} = \left(\frac{1}{x}\right)^n.$$

Therefore

$$\begin{aligned} \frac{d}{dx} x^{-n} &= n \left(\frac{1}{x}\right)^{n-1} \left(-\frac{1}{x^2}\right) \\ &= -n x^{-n-1}. \end{aligned}$$

Writing the same function as $1/x^n$ and applying the reciprocal form of the quotient rule produces the same result.

4.3 Higher Derivatives

The source states: “Higher derivatives are derivatives of derivatives” (printed p. 3; PDF p. 4). A *higher derivative* is found by differentiating again: the second derivative is the derivative of f' , and we can keep going:

$$f'' = (f')', \quad f''' = (f'')', \quad f^{(n)} = (f^{(n-1)})'.$$

Common notations are

$$f'(x) = Df = \frac{df}{dx}, \quad f''(x) = D^2f = \frac{d^2f}{dx^2}, \quad f^{(n)}(x) = D^n f = \frac{d^n f}{dx^n}.$$

The exponent on D counts repeated differentiation; it is not an ordinary power of the value Df .

Repeatedly differentiating x^n reveals a product of descending integers:

$$\begin{aligned} Dx &= 1, \\ D^2x^2 &= D(2x) = 2, \\ D^3x^3 &= D^2(3x^2) = D(6x) = 6, \\ D^4x^4 &= D^3(4x^3) = D^2(12x^2) = D(24x) = 24. \end{aligned}$$

The *factorial* of a positive integer n is

$$n! = n(n-1) \cdots 2 \cdot 1.$$

The observed pattern is therefore $D^n x^n = n!$.

Proposition 4.2. *For every positive integer n ,*

$$D^n x^n = n!.$$

Proof. For $n = 1$, $Dx = 1 = 1!$. Suppose $D^n x^n = n!$. Then the power rule and linearity give

$$\begin{aligned} D^{n+1}x^{n+1} &= D^n(Dx^{n+1}) \\ &= D^n((n+1)x^n) \\ &= (n+1)D^n x^n \\ &= (n+1)n! = (n+1)!. \end{aligned}$$

Thus the result follows by induction. □

Note. The chain rule and higher derivatives answer different questions. The chain rule differentiates one composition; higher-derivative notation records how many times differentiation has been repeated.

Source: [MIT 18.01 Single Variable Calculus](#), Fall 2006, Lecture 4, “Chain Rule, and Higher Derivatives.”